

Chapter Three

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Theory of production

Learning Objective

Define production, input, and output;

Distinguish the differences between short-run and long-run production period;

Define production function;

Explain the concepts of production function with one variable input;

Distinguish the difference between total, average, and marginal product;

Show the relationship between average product and marginal product;

Describe the law of diminishing marginal product;

Identify and analyze the steps of productions;

Explain the concepts of production function with two variable input;

Define Isoquant curve schedule and map;

State the basic characteristics of Isoquant;

Identify the economic region of production; and

Show the effects of technological change on production.

Definition of Production

- Raw materials yield less satisfaction to the consumer by themselves.
- In order to get better utility from raw materials, they must be transformed into outputs.
- However, transforming raw materials into outputs requires inputs such as land, labour, capital and entrepreneurial ability.
- Production is the process of transforming inputs into outputs



- It can also be defined as an act of creating value or utility.
- The end products of the production process are outputs which could be tangible (goods) or intangible (services)

Example

For example, when we get wheat on a plot of land with the help of inputs like labor, capital and seeds, it is termed as production of wheat.

Similarly, when, in a cloth mill, inputs like labor, capital and threads are transformed into cloth, it is called the production of cloth.

Similarly, in an economy, services are also produced. For instance, the services of a teacher, an advocate, a doctor, a singer and servant are also called *production in economics*.

Factors of Production or Inputs

1. **Land:** Refers to all natural resources (land itself, minerals, forests, water, oil, and other natural assets)

- ❑ Limited in supply and Cannot be produced by human effort

2. **Labour:** Represents human effort, both physical and mental, used in the production process.

- ❑ Includes all types of work performed by individuals to create goods and services.

3. **Capital:** Refers to man-made resources used to produce goods and services.

- ❑ Includes machinery, tools, buildings, and technology.

4. **Entrepreneurship:** Involves the ability to organize the other factors of production effectively to create goods and services.

- ❑ Entrepreneurs take risks to innovate, start businesses, and manage resources.

❑ Key Features:

- Combines land, labor, and capital to produce goods or services.
- The reward for entrepreneurship is profit.

- ❑ Example: Startup founders, business owners.

Classification of Raw Materials: Land or working capital

- **Examples of Raw Materials as Part of Land:**
 - **Minerals and Ores:** Iron ore, gold, silver, and other natural mineral deposits.
 - **Forests:** Trees that are part of a forest before they are cut down for wood or paper production.
 - **Soil and Agricultural Land:** Fertile soil that supports farming activities or land used for growing crops.
 - **Water Bodies:** Water in its natural state (e.g., rivers, lakes, or underground aquifers) before being processed or transported.
 - **Oil and Natural Gas:** Crude oil or natural gas in reservoirs beneath the Earth's surface.
 - **Fisheries and Marine Resources:** Fish in the ocean or other aquatic resources before being harvested.
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- **Transition from Land to Working Capital:**
 - **As Part of Land:** While raw materials are still in their natural state (e.g., trees in a forest, oil underground), they are considered part of land.
 - **As Part of Working Capital:** Once these resources are extracted, harvested, or processed (e.g., logs from trees, crude oil in barrels), they become part of the working capital used in production.

Production Function

- The **production function** is purely a technological relationship that expresses the relationship between the output of a good and the different combinations of inputs used in its production.

- A general equation for production function can, for instance, be described as:

$$Q = f(X_1, X_2, X_3, \dots, X_n)$$

- where, Q is output and $X_1, X_2, X_3, \dots, X_n$ are different types of inputs.
- Inputs are commonly classified as fixed inputs or variable inputs.

Fixed and Variable Input

- **Fixed inputs are those inputs** whose quantity cannot readily be changed adjustment in output is required.
- In fact, no input is ever absolutely fixed but may be fixed during an immediate requirement.
- For example, if the demand for Beer rises suddenly in a week, the brewery factories cannot plant additional machinery overnight and respond to the increased demand.
- Buildings, land and machineries are examples of fixed inputs because their quantity cannot be manipulated easily in a short period of time.
- **Variable inputs are those** inputs whose quantity can be altered almost instantaneously in response to desired changes in output.



Period of Production

- **Short-Run:** Short-run refers to the period over which the amount of some inputs, called the 'fixed inputs', cannot be changed.
- For example, the amount of plant and equipment is fixed in the short run.
- The short-run periods of different firms have different durations.
- some firms can change the quantity of all their inputs within a month while it take more than a year for other types of firms.

$$Q_X = f(L, \bar{k})$$

- **Long-Run:** Long-run is defined as the time period during which all factors of production can be varied. A firm can install a new plant or raise a new factory building.

$$Q_X = f(L, K)$$

Production Function with One Variable Input

$$Q_X = f(L, \bar{k})$$

- The **total product (TP)** is the total output amount resulting from different quantities of inputs. If we assume labour (L) as the variable input assuming (capital, etc., held constant).
- **Marginal product of labour (MPL)** is defined as the change in total product (TP) per unit change in variable input, say labour (L), that is,
 - **$MPL = \Delta TP / \Delta L \dots \dots (2)$**
- Similarly, **Average product of labour** may be defined as
 - **$APL = TP / L \dots \dots (3)$**

Hypothetical Schedule of TP, MP and AP and Diagram

Table 4.1: Hypothetical Schedule of TP, MP and AP

Variable Input (L)	Total Product (TP)	Marginal Product (MP)	Average Product (AP)
0	0	—	—
1	5	5	5
2	11	6	5.5
3	18	7	6
4	24	6	6
5	27	3	5.4
6	27	0	4.5
7	25	-2	3.5

The above schedule can also be expressed graphically by drawing TP, MP and AP curves.

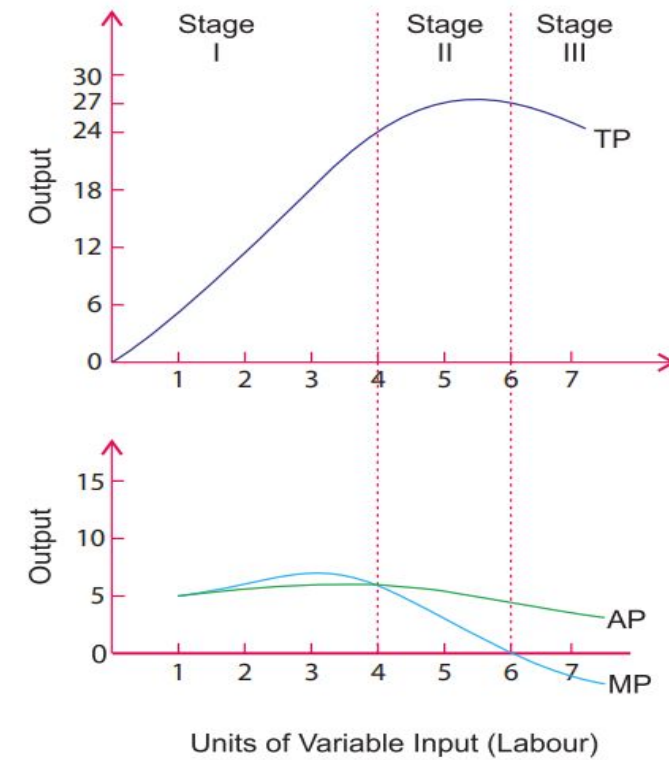
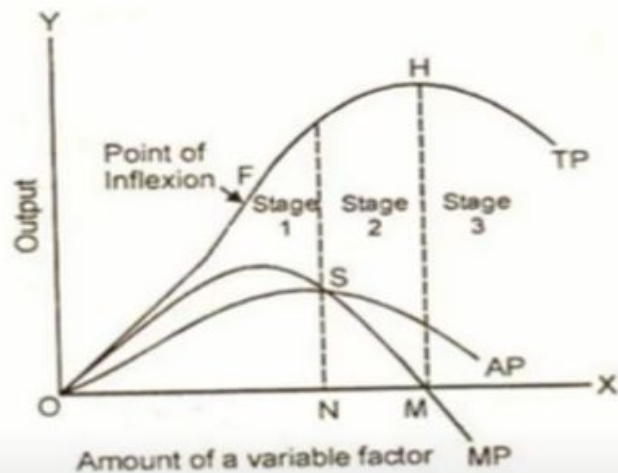


Figure 4.1: TP, MP and AP Curves Showing Three Stages of Production

Stages of Production

The above schedule can also be expressed graphically by drawing TP, MP and AP curves.

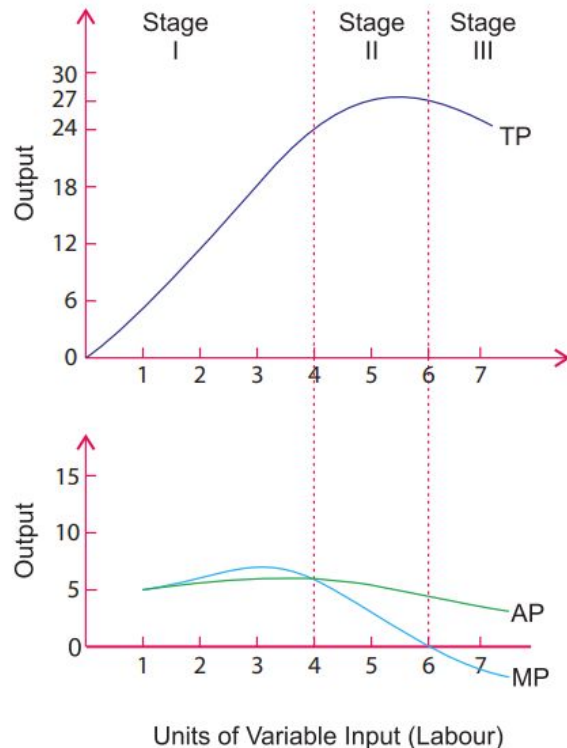


Figure 4.1: TP, MP and AP Curves Showing Three Stages of Production

Table 4.2:

Stage	Total Product	Marginal Product	Average Product
I	Initially increases at an increasing rate and subsequently at a diminishing rate	Increases first, and reaches the maximum	Increases throughout the stage and reaches the maximum
II	Continues to increase at a diminishing rate, and eventually reaches the maximum	Starts to decrease and eventually becomes zero	From the maximum begins to decrease
III	Diminishes	Is negative	Continues to decrease but is always positive

Relationship Between Total Product, Marginal Product, and Average Product

The relationship between MP and AP:

When $MP > AP$, this means that AP is rising,

When $MP = AP$, this means that AP is maximum,

When $MP < AP$, this means that AP is falling.

Graphically, the relationships between the MP curve and AP curve are as follows (see Figure 4.1):

So long as the MP curve lies above the AP curve, the AP curve is a positively sloping curve, AP rises

When the MP curve intersects the AP curve, AP is at maximum,

When the MP curve lies below the AP curve, the AP curve slopes downward, i.e., AP declines.

The relationship between TP and MP:

When TP increases at an increasing rate, marginal product increases,

While TP increases at a diminishing rate, MP declines,

When total product reaches its maximum, marginal product becomes zero,

When TP begins to decline, MP becomes negative.

The Law of Diminishing Marginal Returns

The law states that as more and more of one-factor input is employed, assuming all other input quantities held constant, a point will eventually be reached where additional quantities of the varying input will yield diminishing marginal contributions to total product.

Table 4.3:

Variable Factor (Units of Labour)	Total Product	Marginal Product
1	20	20
2	35	15
3	45	10
4	50	5

The Law of Diminishing Marginal Returns

Note that:

- The law operates only if technology does not change
- The law starts to operate after the MP curve reaches its maximum (see Figure 4.1)
- The law is universal because the tendency of diminishing return is all pervading, and so it applies sooner or later in every field of production.

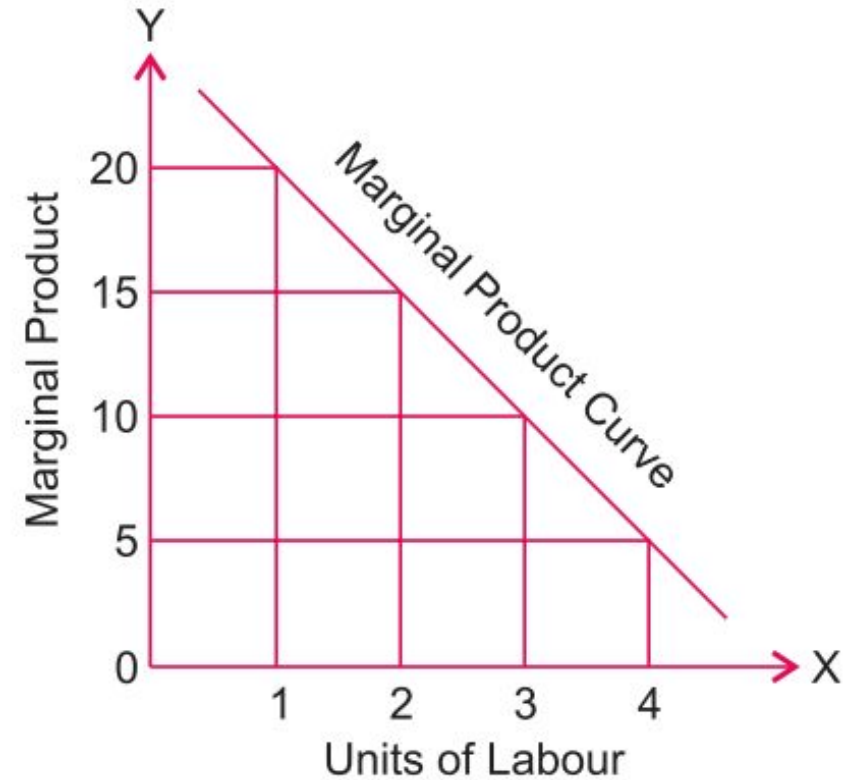


Figure 4.2: Marginal product curve

Long Run Analysis – $Q=f(L,K)$

Production process and iso-quant

	Process P_1	Process P_2	Process P_3
Labour units	$\begin{bmatrix} 2 \\ \end{bmatrix}$	$\begin{bmatrix} 3 \\ \end{bmatrix}$	$\begin{bmatrix} 1 \\ \end{bmatrix}$
Capital units	$\begin{bmatrix} 3 \\ \end{bmatrix}$	$\begin{bmatrix} 2 \\ \end{bmatrix}$	$\begin{bmatrix} 4 \\ \end{bmatrix}$

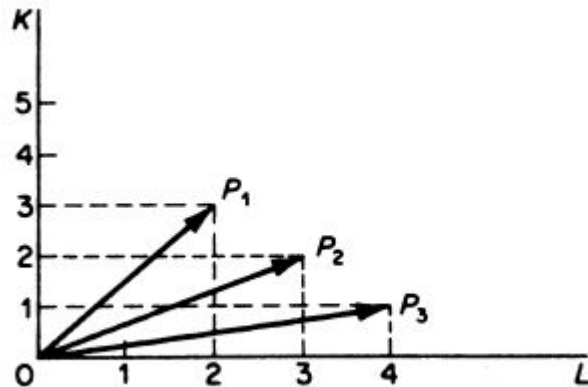


Figure 3.1

• **Technical Efficiency:** Technical efficiency happens when a firm produces a given level of output by using least number of inputs.

• **Economic Efficiency:** Economic efficiency happens when a firm produces a given level of output at least cost. So economically efficient method of production depends on the relative costs of resources.

Production Function with Two Variable Inputs

A tabular representation of the various combinations of two variable inputs which give the same level of output is called an isoquant schedule or equal product schedule.

Table 4.4: Factor Combinations to Produce a Given level of Output (Isoquant Schedule)

Factor Combinations	labour (units)	Capital (units)	Output (units)
A	1	12	20
B	2	8	20
C	3	5	20
D	4	3	20
E	5	2	20

Isoquant curve and Isoquant map

Isoquant: An isoquant is a curve representing the various combinations of two inputs that produce the same amount of output.

An isoquant may, therefore, be defined as a curve which shows the different combinations of the two inputs that produce a given level of output.

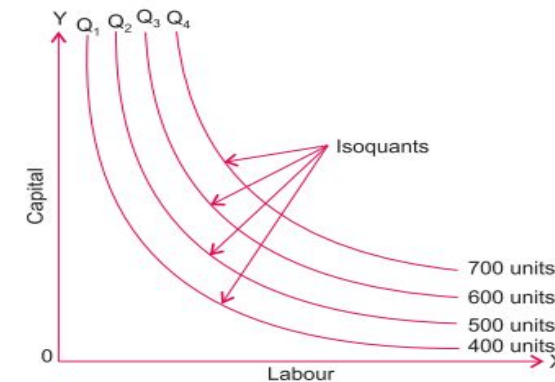
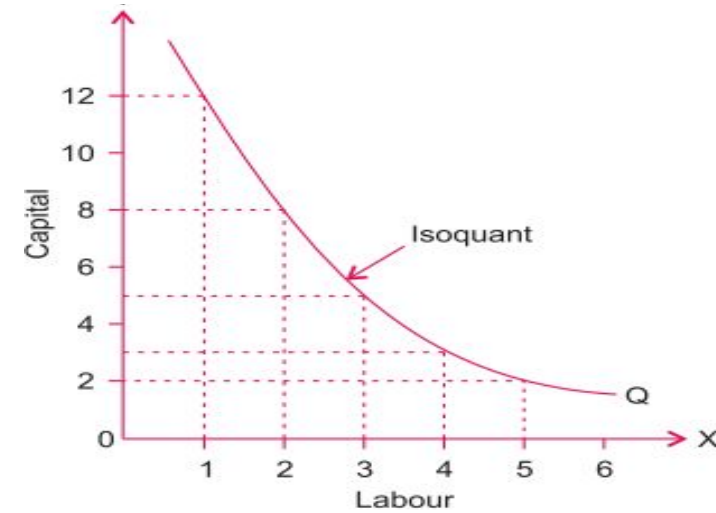


Figure 4.4: Isoquant Map

Properties of Isoquants

- **An isoquant is downward-sloping to the right**, (i.e., negatively inclined), implying that if more of one factor is used, less of the other factor is needed for producing the same level of output.
- **No two isoquants intersect or touch each other.** If two isoquants intersect or touch each other it means that there is a common point on the two curves (point A in Figure 4.6). This common point would imply that the same amount of labour and capital can produce two levels of outputs (example, 60 and 70 units, here), which is impossible.
- **Isoquants are convex to the origin.** The property of convexity implies that the slope of the isoquant diminishes from left to right along the curve. Convexity of an isoquant is the result of the principle of diminishing marginal rate of technical substitution (MRTS) of one factor in place of the other. We discuss MRTS more in a later sub-section.

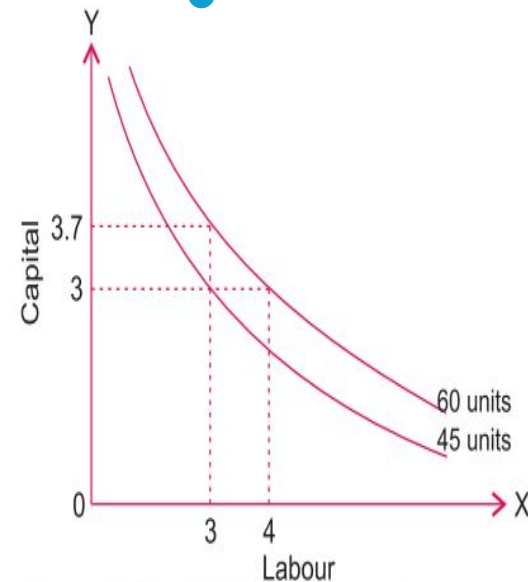


Figure 4.5: Higher Isoquant vs. Lower Isoquant

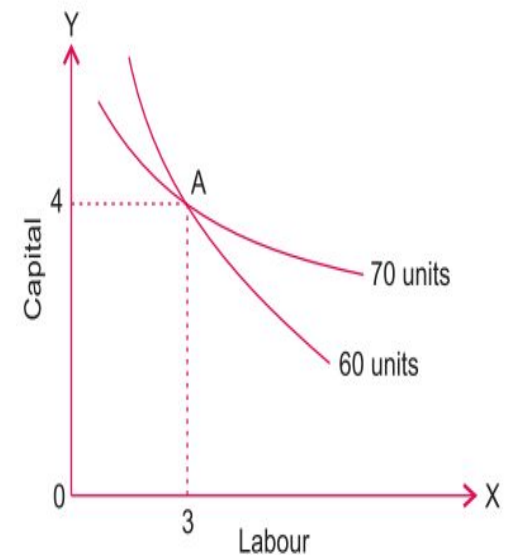


Figure 4.6: Intersection of Isoquants (impossible condition)

Marginal Rate of Technical Substitution (MRTS)

A **marginal rate of technical substitution** is the rate at which factors can be substituted at the margin without altering the level of output.

More precisely, marginal rate of technical substitution of labour for capital may be defined as the number of units of capital which can be replaced by one unit of labour, the level of output remaining unchanged.

Table 4.5: Marginal Rate of Technical Substitution

Factor Combinations	Units of Labour (L)	Units of Capital (K)	MRTS of L for K
A	1	12	—
B	2	8	4
C	3	5	3
D	4	3	2
E	5	2	1

MRTS Formula

- MRTS of labour for capital = $\Delta K / \Delta L$

= Amount of capital given up / Amount of labor used(4)

ΔK represents change in units of capital and ΔL , change in units of labour.

Note:

- MRTS at a point on an isoquant = the slope of the isoquant at that point.
- The property of diminishing MRTS results in convexity of the isoquant.

Economic Region of Production

A **ridge line** is the locus of points of isoquants where marginal product of input is zero.

- the **upper ridge line** joins all such points (for example, a, b, etc.) where marginal product of capital (MP_K) is zero
- **lower ridge line** joins points where marginal product of labour (MPL) is zero (points c, d, etc.).
- The production techniques are technically efficient inside the ridge lines.
- Outside the ridge lines, the marginal products of inputs are negative, i.e., more of both inputs are required to produce a given level of output.
- Obviously, no rational producer would like to operate outside the ridge line. Thus, the economic region of production is the region bounded by the ridge lines.

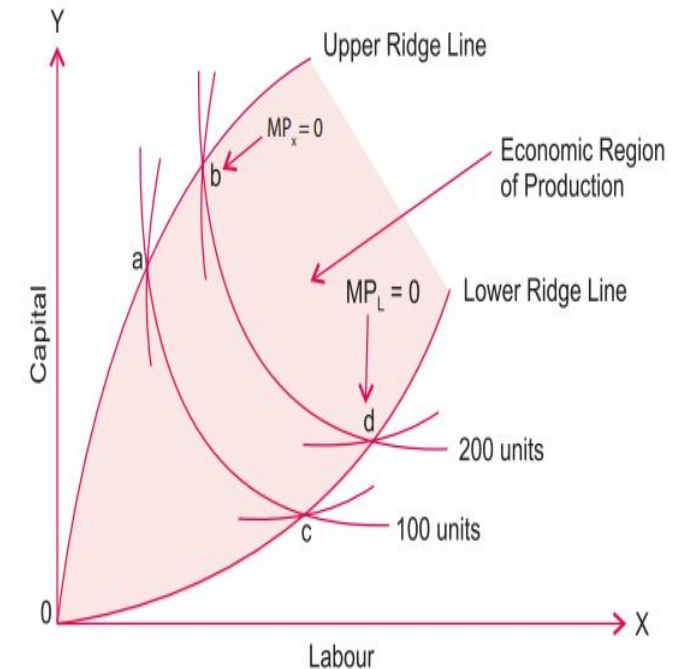


Figure 4.7: Economic Region of Production (Ridge Line)

Returns to Scale (Production with all Variable Inputs)

Returns to Scale (RTS): Returns to scale refers to the rate by which output changes if all inputs are changed by the same proportion.

1. **Increasing Returns to Scale.** It occurs when output increases by a greater proportion than the proportion of increase in all the inputs.
 2. **Constant Returns to Scale.** It happens when output increases by the same proportion as of inputs increase.
 3. **Diminishing Returns to Scale.** It occurs when output increases by a smaller proportion than the proportion of in input increases.
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Returns to Scale and Homogeneity of The Production Function

Suppose we increase both factors of the function : $X_0 = f(L, K)$

by the same proportion t , and we observe the resulting new level of output X^* : $X^* = f(tL, tK)$

- If t can be factored out (that is, may be taken out of the brackets as a common factor), then the new level of output X^* can be expressed as a function of t (to any power v) and the initial level of output

- $X^* = t^v f(L, K)$ or $X^* = t^v x_0$

- and the production function is called homogeneous. If t cannot be factored out, the production function is non-homogeneous. Thus:

- A **homogeneous function** is a function such that if each of the inputs is multiplied by t , then t can be completely factored out of the function. The power v of t is called the degree of homogeneity of the function and is a measure of the returns to scale:

1. If $v = 1$ CRTS . 2. If $v < 1$; DRTS. 3. If $v > 1$; IRTS

Some example

<u>Example of Non- homogeneous production function</u>			<u>Homogeneous Production function with constant returns to scale</u>	
$X_0 = f(L, K) = L + K^2$ $= tL + t^2 K^2$ $= t(L + tK^2)$ $X^* \neq t X_0$			$Q = K^{0.3} L^{0.7}$ $= (tK)^{0.3} (tL)^{0.7}$ $= t^{0.3} K^{0.3} t^{0.7} L^{0.7}$ $= t^{0.3} t^{0.7} (K^{0.3} L^{0.7})$ $= t^1 (K^{0.3} L^{0.7})$ $Q^* = t^1 Q$	
$Q = K^{0.3} L^{0.2}$	$Q = K^2 + L^2$	$Q = 2k^2 + L + KL$	$Q = 2k^2 + L^2 + KL$	$Q = 4k + 4L^2$

Graphical presentation of Returns to scale with isoquants

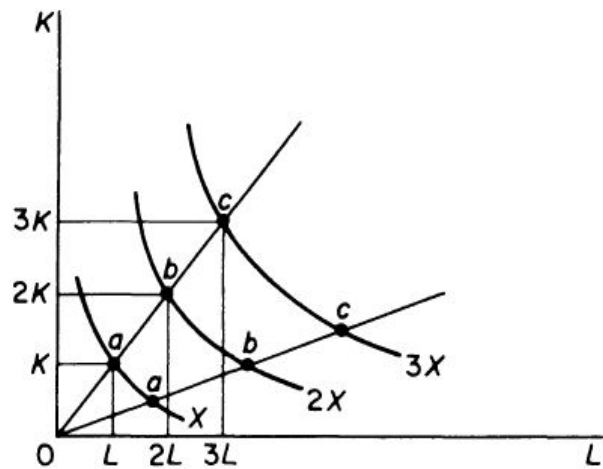


Figure 3.18 Constant returns to scale: $0a = ab = bc$

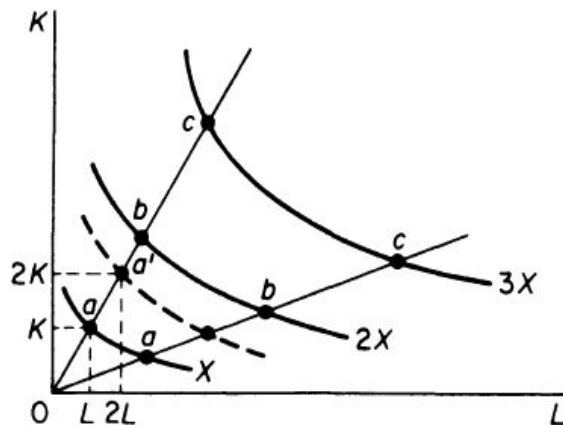


Figure 3.19 Decreasing returns to scale: $0a < ab < bc$

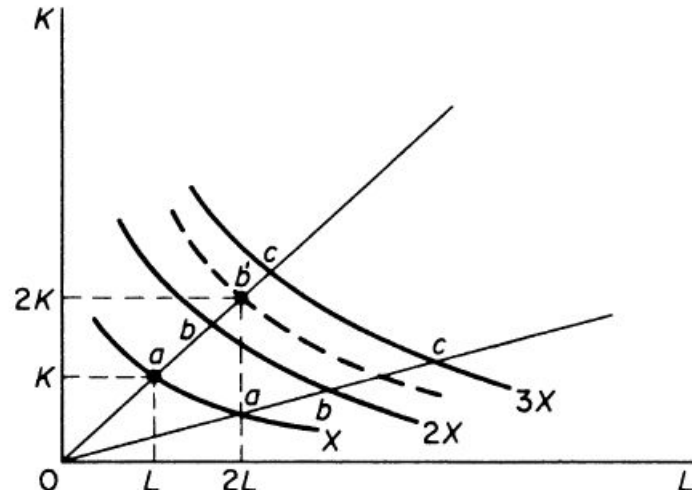
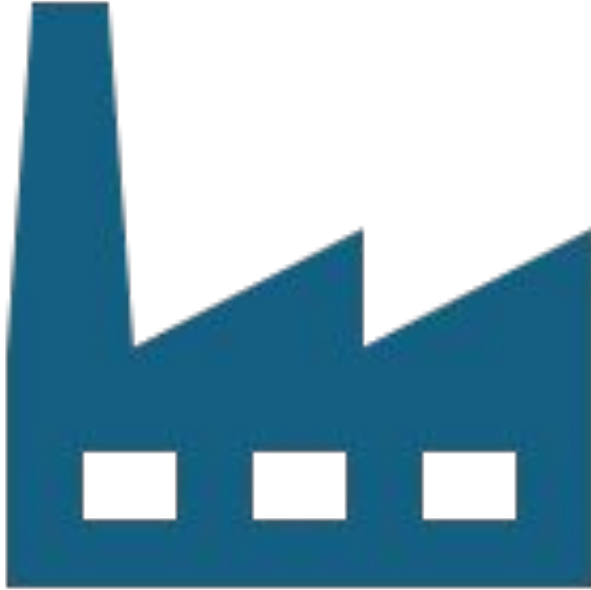


Figure 3.20 Increasing returns to scale: $0a > ab > bc$

Reasons for Increasing and Decreasing Returns



Reasons for operation of **increasing returns to scale** are:

- Greater division of labour and specialisation which increases productivity.
- Use of more productive specialised machinery.

Reasons for operation of **diminishing returns to scale** are:

- The main reason for operation of diminishing return to scale is difficulty in management and coordination when scale of operation becomes bigger and bigger.

Effect of Technological Change on Production Function

- **Technological change** refers to a change in the underlying techniques of production, as occurs when a new process of production is invented or an old process is improved. In such situations, the same output is produced with fewer inputs or more output is produced with the same inputs. These changes in technology are called technological progress or innovation in processes.

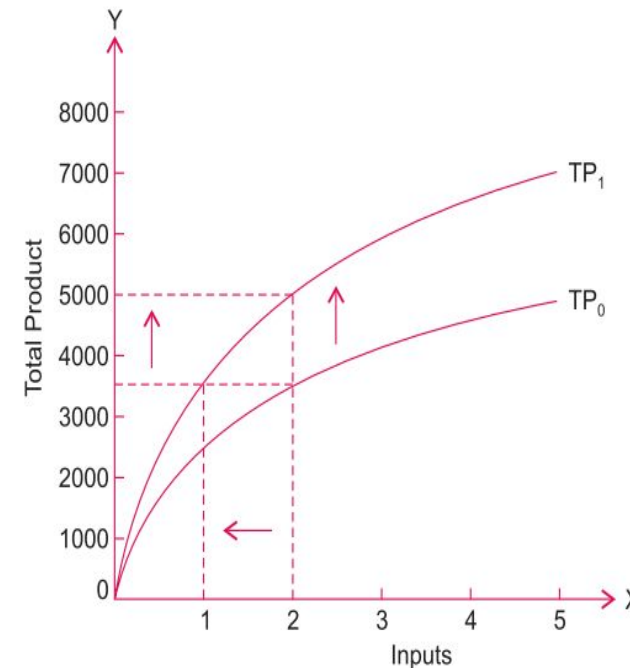


Figure 4.9: Technological Progress Shifts Production Function Upward